Prof. Inese Bula
University of Latvia
Faculty of Physics, Mathematics and Optometry

Lecture Notes «Matemātiskā analīze I»

CALCULUS I

Doc. Līga Zariņa
University of Latvia
Faculty of Computing

Dr. Līga Zariņa
Researcher
University of Latvia
Laboratory for Perceptual and Cognitive Systems
at the Faculty of Computing

Raiņa bulv 19, room 301
E-mail: liga.zarina@lu.lv - please, write to arrange consultations if needed
Phone: +371 28622755

LECTURE 2.

LIMITS I

Intuitive understanding of the limit
Intuitive definition of the limit
Intervals and neighbourhoods
Precise definition of the limit
2. Homework

Limits

Calculus is the study of limits.

The question: What happens to the function f(x) as x gets close to come constant c?

Problems leading to limit concept

- Object steadly moves forward and we know its position at any given time. We denote the position at time t by s(t). How fast is the object moving at time t=1?

We can use: $speed = \frac{distance}{time}$

This is average speed over the interval, but we do not know wheather the speed is constant over the interval. We should consider average speed over smaller and smaller intervals, thus – talk about limit of the average speed.

Limits

- Area of the circle. We cannot calculate a precise area of the circle, but we can calculate the area of a regular polygon with n sies. By taking regular polygon with more and more sides – we are able to approximate the area of a circle to any desired level of accuracy. The area of the circle is the limit of the areas of the inscribed polygons as n increases without bond.

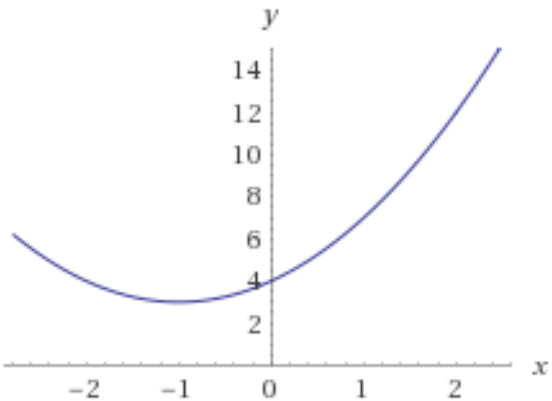
- If consider graph y=f(x) for $a \leq x \leq b$ that is curved. We can find numerous points along the curve and connect them with line segments. If we add up the lengths of these line segments – we should get a sum that is approximately the length of the curve (thai is limit).

Intuitive understanding of the limit

$$f(x) = \frac{x^3 - 8}{x - 2}$$

- Not defined at $x = 2$
- $Dom(f) = (-\infty, 2) \cup (2, +\infty)$

We can, however, still ask what is happening to *f(x)* as *x* approaches 2?



(*x* from −2.8 to 2.5)

x	*f(x)*
2.2	13.24
2.1	12.61
2.01	12.0601
2.001	12.006001
2.0001	12.00060001
↓	↓
2	?
↑	↑
1.9999	11.99940001
1.999	11.994001
1.99	11.9401
1.9	11.41
1.8	10.84

$$\lim_{x \to 2} \frac{x^3 - 8}{x - 2} = 12$$

$$\lim_{x \to 2} \frac{x^3 - 8}{x - 2} = \lim_{x \to 2} \frac{(x-2)(x^2 + 2x + 4)}{(x-2)} = \lim_{x \to 2} (x^2 + 2x + 4) = 2^2 + 4 + 4 = 12$$

This is read «the limit as *x* approaches 1 of (x^3-8)/(*x*-2) is 12»

Intuitive definition of the limit

Lets say that $\lim_{x \to c} f(x) = L$ means that when x is near but different from c then $f(x)$ is near L.

- We do not require anything *at c*. (The function even do not need to be defined *at c*)
- The notion of limit is associated with the behavior of a function ***near c***, not *at c*

What does **near** means? How near is near?

Intervals and neighbourhoods

Interval is a set of real numbers with the property that any number that lies between two numbers in the set is also included in the set.
There are several kinds of intervals.

Open interval – not including the end points ($x: a < x < b$), (a,b),]a,b[
Closed interval – includes the end points ($x: a \leq x \leq b$), [a,b]

If $a < b$

$(-\infty; a) = \{x \mid x \in \mathbb{R} \,\&\, x < a\}$

$(-\infty; a] = \{x \mid x \in \&\mathbb{R} \leq x, a\}$

$(a; b) = \{x \mid x \in \mathbb{R} \,\&\, a < x < b\}$

$[a; b) = \{x \mid x \in \mathbb{R} \,\&\, a \leq x < b\}$

$(a; b] = \{x \mid x \in \mathbb{R} \&\ a < x \leq b\}$

$[a; b] = \{x \mid x \in \mathbb{R} \,\&\, a \leq x \leq b\}$

$[b; +\infty) = \{x \mid x \in \mathbb{R} \,\&\, b \leq x\}$

$(b; -\infty) = \{x \mid x \in \mathbb{R} \,\&\, b < x\}$

$\mathbb{R} = (-\infty; +\infty)$

Intervals and neighbourhoods

We follow the tradition in using Greek letters ε (epsilon) and δ (delta) to stand for (usually small) arbitrary positive numbers.

The set $\mathfrak{A}(x_0, \delta) = (x_0 - \delta; x_0 + \delta) = \{x | |x - x_0| < \delta\}$ is called $\delta - neighbourhood$ of the x_0.

A deleted neighbourhood of a point x_0 (sometimes called a punctured neighbourhood) is a neighbourhood of p, without x_0.

The set $\dot{\mathfrak{A}}(x_0, \delta) = (x_0 - \delta; x_0 + \delta)\{x_0\} = \mathfrak{A}(x_0, \delta)\{x_0\} = \{x | 0 < |x - x_0| < \delta\}$ is called deleted $\delta - neighbourhood$ of the x_0.

The concept of deleted neighbourhood occurs in the definition of the limit of a function.

Precise definition of the limit

To say that $\lim_{x \to c} f(x) = L$ («The limit as *x* approaches c of f(x) is L») means that for each given $\varepsilon > 0$ (no matter how small) there is a corresponding $\delta > 0$ such that $|f(x) - L| < \varepsilon$, provided that $0 < |x - \mathrm{c}| < \delta$; that is,

$$0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon$$

or

$$x \in \dot{\mathfrak{U}}(c, \delta) \Rightarrow f(x) \in \mathfrak{U}(L, \varepsilon).$$

(for each $x \in Dom(f)$)

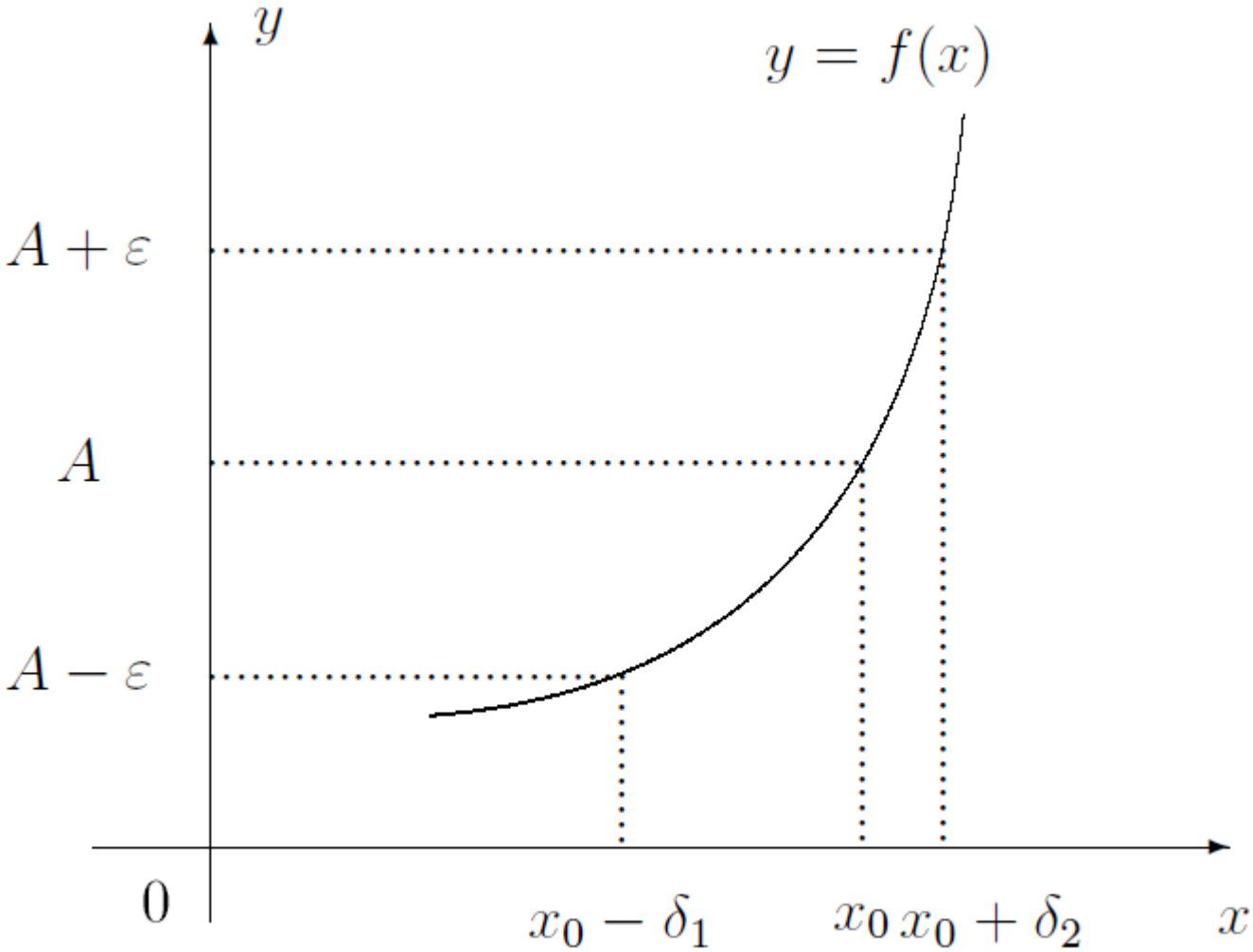
(«The limit as *x* approaches c of f(x) is L» $(\lim_{x \to c} f(x) = L)$) $\Leftrightarrow$

$(\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in Dom(f)[0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon])$.

Precise definition of the limit

$$\lim_{x \to c} f(x) = L \;\Leftrightarrow$$

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in Dom(f)[0 < |x - x_0| < \delta \Rightarrow |f(x) - A| < \varepsilon]$$



$$\delta \leq \min\{\delta_1; \delta_2\}$$

Precise definition of the limit

$\lim_{x \to c} f(x) = L \iff$
$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in Dom(f)[0 < |x - x_0| < \delta \Rightarrow |f(x) - A| < \varepsilon]$

Example

If $f(x) = c \in \mathbb{R}$
then $\lim_{x \to x_0} f(x) = c$

This example is atypical, because we do not need to care about $\varepsilon > 0$.

$|f(x) - c| = |c - c| = 0$,
and it is smaller than any positive ε.

- for δ we can take any positive number

Precise definition of the limit

Example

Proof that $\lim\limits_{x\to 2}(3x+2)=8$.

We have to proof that

$\forall\varepsilon>0\ \exists\delta>0\ \forall x\in\mathbb{R}[0<|x-2|<\delta\Rightarrow|3x+2-8|<\varepsilon]$.

(Let ε be any positive number. We must produce δ such that

$0<|x-2|<\delta\Rightarrow|3x+2-8|<\varepsilon$)

Consider the inequality on the right (transform by using equivalent inequalities).

$$|3x+2-8|<\varepsilon,$$
$$|3x-6|<\varepsilon,$$
$$3|x-2|<\varepsilon,$$
$$|x-2|<\frac{\varepsilon}{3}.$$

From this follows: if $0<\delta\leq\frac{\varepsilon}{3}$, then $|3x+2-8|<\varepsilon$

Now we see how to choose δ, that is $\delta=\frac{\varepsilon}{3}$. Of course any smaller δ would work.

We have prooved that

$\forall\varepsilon>0\ \exists 0<\delta=\delta(\varepsilon)=\frac{\varepsilon}{3}\ [|x-2|<\delta\Rightarrow|3x+2-8|<\varepsilon]$.

Precise definition of the limit

The proof can be divided in two parts:
1) Analysis to find valid δ
2) Demonstration that the selected δ meets all requirements of the limit definition

Usually from the analysis part the scheme of the proof is clear and so it is mostly limited to analysis.

In order to avoid complicated algebraic evaluations, there is a limit theory, which we will talk about in the next lecture.

Precise definition of the limit
One-sided limits

Right-hand limit

To say that $\lim_{x \to c^+} f(x) = L$ means that for each $\varepsilon > 0$ there is a corresponding $\delta > 0$ such that

$$0 < x - c < \delta \Rightarrow |f(x) - L| < \varepsilon$$

$(\lim_{x \to c^+} f(x) = L) \Leftrightarrow$
$(\forall \varepsilon > 0 \exists \delta > 0 \forall x \in Dom(f)[0 < x - c < \delta \Rightarrow |f(x) - L| < \varepsilon])$

Left-hand limit

To say that $\lim_{x \to c^-} f(x) = L$ means that for each $\varepsilon > 0$ there is a corresponding $\delta > 0$ such that

$$0 < c - x < \delta \Rightarrow |f(x) - L| < \varepsilon$$

$(\lim_{x \to c^-} f(x) = L) \Leftrightarrow$
$(\forall \varepsilon > 0 \exists \delta > 0 \forall x \in Dom(f)[0 < c - x < \delta \Rightarrow |f(x) - L| < \varepsilon])$

Precise definition of the limit

Example

$$sgn(x) = \begin{cases} -1, x < 0 \\ 0, x = 0 \\ 1, x > 0 \end{cases}$$

We are interested in point x=0.

$$\lim_{x \to 0^+} sgn(x) = 1 \; and \; \lim_{x \to 0^-} sgn(x) = -1,$$

but $sgn(0) = 0$.

Theorem

$$(\lim_{x \to c} f(x) = L) \Leftrightarrow (\lim_{x \to c^-} f(x) = L = \lim_{x \to c^+} f(x))$$

Precise definition of the limit

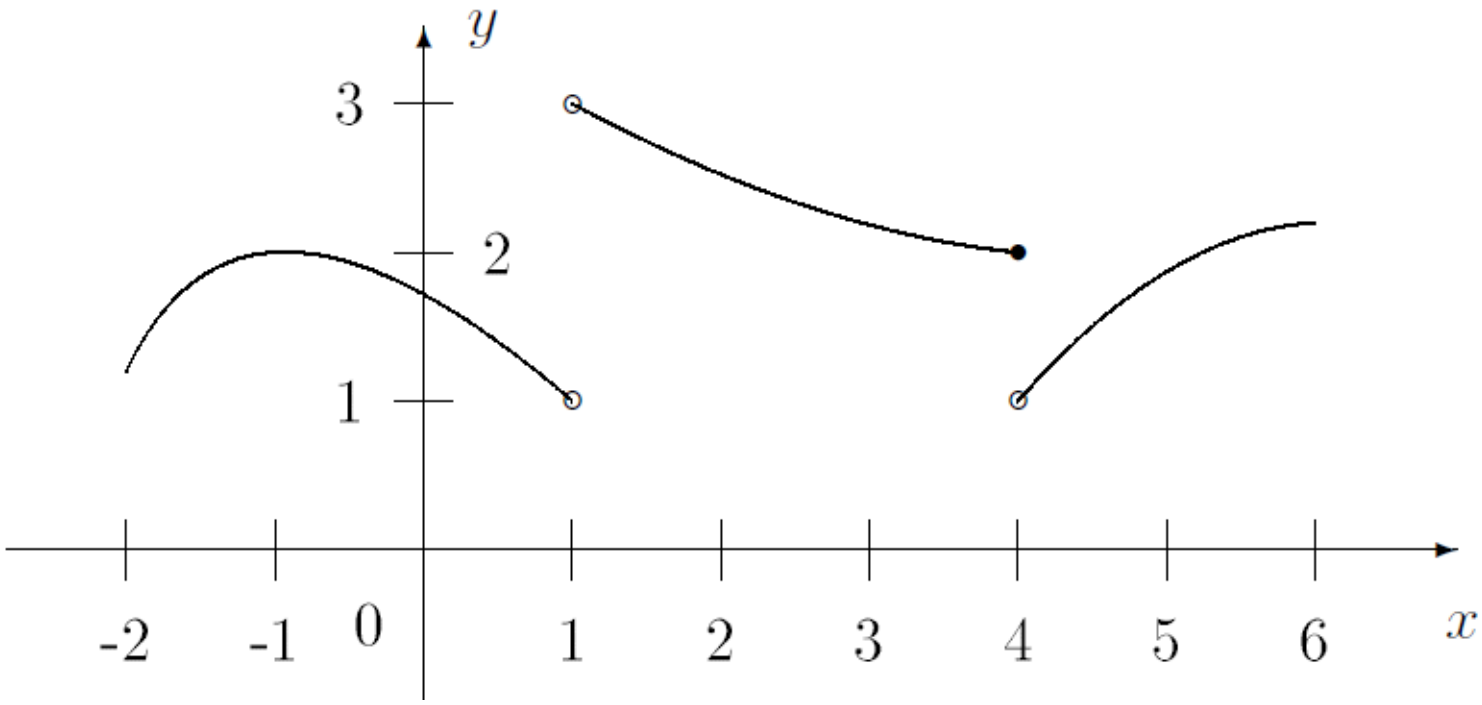
Example

We have grapg of function *f(x)*.

Empty dots – do not belong to the graph, fille dots – belong to the graph.

Find:

$$\lim_{x\to -1} f(x) \quad \lim_{x\to 1^-} f(x) \quad \lim_{x\to 1^+} f(x) \quad \lim_{x\to 4} f(x) \quad \lim_{x\to 4^-} f(x) \quad \lim_{x\to 4^+} f(x)$$



HOMEWORK 2

3. Please, in accordance with the limit definition proof that

$$\lim_{x\to 4}\frac{2x^2-5x-12}{x-4}=11$$

Answer: $0<\delta\le\frac{\varepsilon}{2}$

4. Please, sketch the graph of the function: $f(x)=\begin{cases} x^2,\ x\le 0 \\ x,\ 0<x<1 \\ 1+x^2,\ x\ge 1\end{cases}$,

And find the limits if they exist:

a) $\lim\limits_{x\to 0} f(x)$, b) $\lim\limits_{x\to 1} f(x)$, c) $\lim\limits_{x\to 1^-} f(x)$, d) $\lim\limits_{x\to 1^+} f(x)$

Reading: D. Varberg, E. J. Purcell, S. E. Rigdon *Calculus*. 9th Edition, Prentice-Hall, Inc.m, 2014, p.55 – 67.

Chapter 1. Sections 1.1. and 1.2.